EYEQ: A Hybrid Quantum–Classical Eye-Gaze Typing System on Commodity Webcams

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Abstract—Text entry still assumes working hands and a physical keyboard, excluding users with motor impairments and failing in hands-occupied or sterile settings. We present an eye-gaze typing system that runs entirely on a commodity webcam, with no infrared hardware and no client build step. Gaze is estimated by mapping MediaPipe iris landmarks through a per-user ridge-regression calibration, and characters are committed by dwell selection over either a per-letter keyboard or large, gaze-friendly key clusters. Clustering makes targets easy to hit but renders the input ambiguous: several letters share one region, so distinct words collapse onto the same cluster code. To resolve that ambiguity accurately and in real time we introduce a quantum conditional selection rule. Candidate targets form a superposition whose amplitudes carry the gaze likelihood; a language-conditioned controlled unitary fuses linguistic context through both amplitude and phase; and a Born-rule measurement commits the most probable candidate. Unlike classical Bayesian fusion, which multiplies marginals, the phase term admits interference between context-coherent hypotheses. We prove the rule is strictly more expressive — it recovers log-linear fusion exactly at zero phase, and no classical mixture reproduces its cross-terms — and a consistency-preserving oracle guarantees that every decoded output re-encodes to exactly the gazed clusters, never an off-target word. The rule is realised as a lightweight classical state-vector simulation over a five-qubit register, well inside the webcam frame budget. We then measure it, and report that the added expressiveness does not pay. At matched prompt rate the interference term is worth a median -0.03 percentage points across five noise levels, with no significant win or loss; amplitude amplification lengthens mean dwell from 2.63 to 2.80 frames and costs two points of commit accuracy. We give the mechanism rather than the shrug: coherence is anti-correlated with correctness, because the correct letter after an English prefix carries the richest hypothesis support and therefore the widest spread of phases. The accuracy instead comes from the front end. Replacing eyelid-relative normalisation of the iris offset with a canthus-relative one reduces held-out error from 5.55° to 3.76°, a 32.3 % reduction measured across nine participants over eleven sessions. The effect improves every participant without exception ($t(8) = 9.31$, $p = 1.4 \times 10^{-5}$, $d_z = 3.10$), subsumes explicit head pose, and is invariant to head roll by construction to 10^{-15}.

Index Terms—Quantum conditional selection rule; quantum-inspired inference; interference-based gaze–language fusion; amplitude amplification; cluster-constrained decoding; eye-gaze typing; webcam gaze estimation; dwell-based selection; assistive text entry; facial landmarks; calibration.

I. Introduction

Eye typing offers people with severe motor impairment a channel for written communication that requires no limb movement. Commercial infrared eye trackers achieve roughly 0.25–0.5° of gaze error but cost substantially more than the computers they attach to, so a large body of work has pursued the same capability using only a consumer webcam.

A webcam gaze keyboard faces two problems in series, and they are not the same kind of problem. The first is estimation: turning a low-resolution, uncalibrated image of a face into a screen coordinate. The second is decision: turning a noisy coordinate and a language context into a committed character. Most of the literature addresses the first and treats the second as a hit test.

This paper addresses both, and finds the balance between them counter-intuitive.

For the decision problem we introduce a quantum conditional selection rule. The deployed system already fuses gaze and language the conventional way, by multiplying a gaze posterior with a language prior. That product is a product of marginals, and a marginal has already discarded the structure that produced it: whether a letter's linguistic support came from one confident word or from fifty diffuse ones. The selection rule keeps the hypotheses distinct, places them in superposition with amplitudes carrying the gaze likelihood, applies a language-conditioned unitary that attaches both a magnitude and a phase

to each, and reads the result with the Born rule. The phase channel admits interference, which no product of marginals can express.

That is a genuine generalisation and we prove it. It is also, we find, an empirically empty one, and the more useful contribution of this paper is the explanation of why. The interference term is systematically mis-signed for this task: the correct letter after an English prefix is usually the one with the richest vocabulary support, so it carries the most hypotheses, the widest spread of phases, and hence the lowest coherence. Phase derived from hypothesis likelihood penalises exactly the evidence it should reward, and no retuning of the phase function repairs the sign.

For the estimation problem the literature has converged on an evaluation habit that, we argue, obscures progress. Systems are reported as classifiers: the metric is the proportion of frames in which the gaze direction is assigned to the correct region, where the regions number between two and nine. Reported figures of 85–93 % are common. This is a legitimate measurement of a legitimate design — most such systems select keys by blinking or by scanning, so a coarse direction is all they require — but it is not comparable to the continuous angular error that the wider gaze literature reports, and it cannot tell a reader whether a user could point at a key.

We take the harder formulation: a continuous screen coordinate (Section IV), evaluated on targets never used for fitting under the protocol of Section VI, and reported in degrees of visual angle. Working in that regime surfaced a defect in a step the literature treats as settled, and it is there — not in the decision rule — that the accuracy turned out to be.

A. Contributions

1) A quantum conditional selection rule for gaze–language fusion: candidates in superposition with gaze-derived amplitudes, a language-conditioned controlled unitary carrying context in amplitude and phase, a consistency-preserving oracle, and Born-rule commitment, realised as classical state-vector simulation over a five-qubit register.
2) A containment proposition. The rule reduces to the deployed log-linear fusion exactly, to floating point, at zero phase (Proposition 1), and its cross-terms are unreachable by any classical mixture of the same hypotheses (Proposition 2). Containment is what makes the head-to-head comparison honest rather than rigged.
3) A measured null with a mechanism. Interference is worth a median -0.03 percentage points at matched prompt rate and amplification lengthens dwell. We trace this to a sign: the measured coherence separation between correct and incorrect letters is negative at every phase scale and worsens monotonically as the scale grows.
4) Canthus-relative normalisation. Replacing eyelid-relative normalisation of the iris offset with a canthus-relative one reduces held-out error by 32.3 % ($5.55^\circ \rightarrow 3.76^\circ$), improving all nine participants without exception. The mechanism is anatomical: the eyelid moves independently of gaze, the canthi do not. To our knowledge this normalisation appears in none of the surveyed works.
5) An explanation for a reported but undiagnosed asymmetry. One surveyed system reports 93 % horizontal against 76 % vertical accuracy and does not account for the gap. Eyelid contamination predicts exactly this asymmetry, since only the vertical channel is normalised by the aperture.
6) Roll invariance by construction. Projecting the offset onto the canthus axis and its perpendicular removes head-roll dependence without estimating the roll angle. We verify invariance numerically to $\sim 10^{-15}$ across 0–40°.
7) Reported negative results, including quantum-kernel regression, explicit head-pose features, an iris-radius scale cue, eye-region de-rotation and discrete hierarchical selection — each implemented, measured, rejected, and recorded so the next group need not repay the cost.

II. Related Work

Table I summarises ten recent works on webcam gaze keyboards and related interfaces. We group them by how the iris is localised, since that choice determines what the rest of the system can do.

Threshold and projection methods. Saraswati et al. [8] detect the face with a Haar cascade and locate the iris by binary thresholding followed by integral projection. The approach is inexpensive but the threshold must be retuned for each lighting condition — the paper tabulates separate values for 8, 30 and 81 lx — and detection fails beyond 50 cm.

Geometric circle fitting. Cabatuan et al. [9] treat the iris as a circle-detection problem, applying Sobel edge detection, a thinning step and a Circular Hough Transform, and implement the pipeline on an FPGA in Verilog. It is the only surveyed work to target dedicated hardware.

Landmark regression with aspect ratios. The largest group [2], [3], [5], [7] uses the 68-point dlib shape predictor and derives an Eye Aspect Ratio (EAR) or a horizontal/vertical gaze ratio. A property of this family deserves emphasis: the 68-point model contains no iris landmarks at all. It locates the eye corners and lids, and everything concerning the pupil is inferred from those. This is why all four of these systems select keys by blinking or scanning rather than by pointing — the representation cannot support a continuous coordinate.

Dedicated iris landmarks. Rafid et al. [4] and the virtual-mouse system of [6] use MediaPipe Face Mesh, which supplies a five-point iris ring per eye in addition to 468

face landmarks. This yields the iris centre directly and is the prerequisite for continuous regression. Only two of the ten surveyed works adopt it, both dated 2023 or later.

End-to-end learning. Waideman and Aquino Junior [1] regress gaze from pixels with a four-stream CNN and are the only surveyed work to model head pose, supplying head position as a fourth network channel. Abhaya et al. [3] classify nine gaze directions with a CNN.

Selection strategy. Zhou et al. [10] address the Midas touch problem with an adaptive dwell time that shortens as the user's input speed rises, using a commercial SMI eye tracker rather than a webcam.

Ambiguous input and language models. Enlarging targets to make them easy to hit, then recovering the intended word with a language model, is the established trade in text entry: SHARK^2 [18] decodes whole-word gestures against a lexicon, and the same bargain underlies our cluster layout. Our addition is the constraint direction. Where a conventional decoder scores hypotheses and may return a word the input does not support, the consistency oracle of Section V admits only hypotheses that re-encode to exactly the gazed sequence.

A. Positioning

Two observations follow from Table I and motivate this work.

First, no surveyed work reports gaze error in degrees. The reported percentages measure region classification, often over as few as two or three zones where chance alone is 33–50 %. Our results therefore cannot be ranked against them, and we do not attempt to; we state this explicitly because a table of mixed metrics invites exactly that error.

Second, head pose is almost entirely absent. One work models it implicitly [1]; one prohibits head movement and reports the resulting discomfort as a limitation [2]; one observes that head shaking causes error and does not address it [10]. No surveyed work performs geometric head-pose estimation. Section VII offers a partial explanation for why that omission has cost less than might be expected.

III. The Vertical-Asymmetry Observation

The observation that motivates our method is available in the surveyed literature but is not commented upon there. Waideman and Aquino Junior [1] report directional accuracy split by axis: 93 % for left/right and 76 % for up/down. The gap is large and the paper does not explain it.

Consider how the vertical channel is conventionally formed. Let $\mathbf{p}$ be the iris centre and let t, b be the upper and lower eyelid landmarks. The standard vertical feature is

$$v = \frac{p_y - t_y}{b_y - t_y + \epsilon}, \tag{1}$$

that is, the iris height expressed as a fraction of the eyelid aperture. The denominator of (1) is not a fixed anatomical quantity. It contracts with every squint, every partial blink, and with ambient brightness. Any such change alters v without the eye having moved. The horizontal channel, normalised by the distance between the eye corners, suffers no equivalent contamination.

An eyelid-normalised vertical channel should therefore be measurably worse than its horizontal counterpart — which is what [1] reports. Our method follows directly: normalise against a structure that does not move.

IV. Front End: Estimating the Coordinate

A. Landmark acquisition

We use MediaPipe Face Landmarker [11], which returns 478 landmarks including a five-point iris ring per eye (indices 468–472 and 473–477) and a 4×4 facial transformation matrix. Detection runs on the GPU where available and falls back to CPU/WASM otherwise. The medial and lateral canthi are landmarks 133/33 (left) and 362/263 (right).

B. Canthus-relative normalisation

Let $\mathbf{i}$, $\mathbf{m}$ and $\mathbf{l}$ denote the iris centre and the medial and lateral canthi of one eye. We define the eye axis and its perpendicular,

$$\mathbf{u} = \frac{\mathbf{m} - \mathbf{l}}{\|\mathbf{m} - \mathbf{l}\|}, \qquad \mathbf{u}^{\perp} = \begin{bmatrix} -u_y \\ u_x \end{bmatrix}, \tag{2}$$

and take the iris offset from the canthus midpoint, $\mathbf{d} = \mathbf{i} - \frac{1}{2}(\mathbf{m}+\mathbf{l})$. The two features per eye are the components of that offset in the eye's own frame, scaled by the inter-canthal distance $w = \|\mathbf{m} - \mathbf{l}\|$:

$$c_{\parallel} = \frac{\mathbf{d} \cdot \mathbf{u}}{w}, \qquad c_{\perp} = \frac{\mathbf{d} \cdot \mathbf{u}^{\perp}}{w}. \tag{3}$$

Two properties follow. The scale w is a Euclidean distance rather than a coordinate difference, so it does not collapse as the head turns. And because $\mathbf{u}$ rotates with the head, (3) is invariant to head roll by construction — no roll angle is estimated, and none is needed. Fig. 1 contrasts the two normalisations.

C. Feature vector and calibration

The full vector concatenates the four conventional ratios, head translation and scale proxies, a bias term, and the four canthus components of (3), giving $d = 12$. Screen coordinates are obtained by ridge regression,

$$\mathbf{W} = (\mathbf{F}^{\top}\mathbf{F} + \lambda \mathbf{I})^{-1}\mathbf{F}^{\top}\mathbf{T}, \tag{4}$$

fitted to a short per-user calibration of nine on-screen targets anchored to key centres, five fixations each. Samples are weighted by a geometric frame-quality score; ordinary refinement clicks gathered during use are admitted at a tenth of the weight of a calibration click, since they are not verified ground truth.

TABLE I
Survey of Ten Recent Webcam Gaze-Keyboard Systems. The Metric Column is the Point of the Table: Reported Figures Measure Region Classification, Not Continuous Gaze Error, and Are Therefore Not Comparable With This Work.

Work	Face	Iris localisation	Selection	Head pose	Reported
Waideman et al. [1]	MMOD-CNN	4-stream CNN (face, both eyes, head)	CNN regression	Yes, 4th channel	93 % L/R, 76 % U/D
Islam et al. [2]	HOG + SVM	dlib 68-pt, midpoint	Blink, sequential scan	Movement prohibited	CPM/WPM only
Abhaya et al. [3]	dlib	dlib 68-pt landmarks	CNN, 9 directions (T9)	No	15.3 % error
Rafid et al. [4]	MediaPipe	MediaPipe iris	2×4 grid, blink	No	14.53 CPM
Varshini et al. [5]	dlib	dlib 68-pt, EAR	Blink, group scan	No	92.5 %, 17 WPM
Virtual mouse [6]	MediaPipe	MediaPipe Face Mesh iris	Blink duration	Cited, not implemented	—
Kathpal et al. [7]	OpenCV/dlib	Horizontal/vertical gaze ratio	Blink	No	91 % / 85 % (glasses)
Saraswati et al. [8]	Haar cascade	Threshold + integral projection	Pointer, blink	No	40–50 cm only
Cabatuan et al. [9]	—	Sobel + thinning + Circular Hough	On-screen keyboard	No	FPGA ≡ MATLAB
Zhou et al. [10]	—	SMI ETG 2w (commercial IR)	Adaptive dwell, 600 ms	Noted, unsolved	—
This work	MediaPipe	MediaPipe iris, canthus-normalised	Dwell pointing	Implicit (canthus frame)	**3.76**° continuous

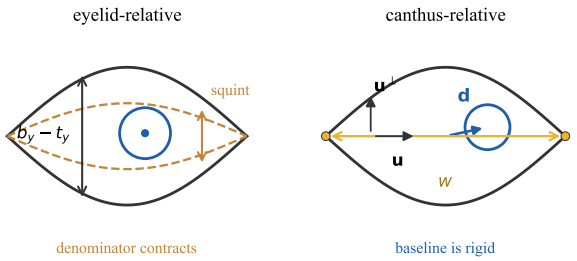


Fig. 1. Eyelid-relative against canthus-relative normalisation of the iris offset. Only the latter is invariant to eyelid aperture and to head roll.

D. Components integrated around the estimator

The coordinate a regressor emits is not yet a usable pointer, and several components between the two are load-bearing. We list them because they are the difference between the numbers in Section VII and a system that is unpleasant to use, and because two of them were added in response to measured defects rather than to intuition.

Temporal filtering. Predictions pass through a One Euro filter [12], whose cutoff rises with speed so that a fixation is smoothed heavily while a saccade is not lagged. Within the dwell window the committed point is the fixation centroid rather than the instantaneous estimate.

Dwell hysteresis. A cursor resting on a key boundary would otherwise oscillate between neighbours and commit neither. The dwell target is retained until the estimate leaves the key by a fixed margin for several consecutive frames, which converts boundary jitter into a stable selection.

Coverage-aware calibration eviction. The calibration buffer is bounded. Evicting oldest-first appears harmless and is not: it silently destroys spatial coverage, and we measured a buffer that had collapsed from thirteen occupied grid cells to two, with error rising from 4.93° to 8.25°. Eviction is therefore driven by grid occupancy, the original calibration anchors are evicted last, and unverified refinement samples enter at a tenth of the weight of a calibration click.

Geometric quality gating. Samples are weighted by a frame-quality score combining face scale, centring and landmark stability, and an adaptive eye aspect ratio baseline suppresses frames during blinks. The baseline is per-user because absolute EAR varies more between people than a blink varies within one.

User-centred pose convention. The facial transformation matrix is right-handed in the camera's frame, which inverts the sense of yaw and pitch relative to how a user describes their own movement. Uncorrected, a head-movement recording is labelled backwards. Correcting the signs does not improve accuracy — a sign flip is absorbed by the regression weights — but it is a prerequisite for any posture-stratified analysis being meaningful.

Reliability gating. A movement classifier refuses to commit from windows that look like saccades or blinks, and a Bayesian network converts its state estimate into $P(\text{key correct})$, which tempers the fused posterior so that a marginal frame falls below the commit threshold and asks instead. A watchdog disables the gate after consecutive refusals, since a gate whose input drifts out of distribution can otherwise refuse every keystroke.

Capture controls. Exposure and contrast are exposed in the interface, because the illumination range over which landmark detection is stable is narrower than the range a webcam will silently auto-expose across.

V. A Quantum Conditional Selection Rule

A. What a product of marginals discards

At commit time the deployed system scores each candidate key k by multiplying the gaze posterior with a language prior,

$$\text{score}(k) \;=\; P(k \mid \text{gaze}) \cdot \max\big(P(k \mid \text{prefix}),\, \varepsilon\big)^{\beta}, \qquad (5)$$

with $\beta < 1$ damping the prior so that the language model advises rather than vetoes, and ε flooring it so that no letter becomes untypable. $P(k \mid \text{gaze})$ is the exact integral of the gaze Gaussian over the key rectangle; $P(k \mid \text{prefix})$ comes from a frequency-weighted prefix trie backed off to a character n-gram.

The second factor is a marginal. It is the total frequency mass of every vocabulary word that continues the prefix with k, and forming that sum destroys the distinction between a letter supported by one dominant continuation and one supported by a diffuse spray that agrees on nothing beyond a single character. After the prefix th, the letters e and r may carry comparable marginal mass while meaning very different things. Equation (5) cannot tell them apart, because the information was gone before the product was taken.

B. The rule

Let H be the vocabulary completions of the current prefix, each with frequency weight w_h and each proposing a next letter k_h. Candidates occupy a five-qubit register spanning a–z and a word-boundary symbol, prepared in superposition with amplitudes carrying the gaze likelihood:

$$|\psi\rangle \;=\; \sum_{k} \sqrt{P(k \mid \text{gaze})}\; |k\rangle. \qquad (6)$$

A language-conditioned diagonal unitary then attaches to each candidate both a magnitude and a phase. Writing $v_h = w_h / P(k \mid \text{prefix})$ for the hypothesis weights within letter k, so that $\sum_h v_h = 1$, the conditioned amplitude is

$$a(k) \;=\; \sqrt{P(k \mid \text{gaze})}\; P(k \mid \text{prefix})^{\beta/2} \underbrace{\sum_{h:\, k_h = k} v_h\, e^{i\varphi_h}}_{c(k)}, \qquad (7)$$

and the Born rule commits the most probable candidate, $P(k) = |a(k)|^2$. The factor $c(k)$ is the coherence of letter k: a weighted mean phasor, with

$$|c(k)|^2 = \sum_{h} \sum_{h'} v_h v_{h'} \cos(\varphi_h - \varphi_{h'}), \qquad (8)$$

whose off-diagonal terms are the interference. Since $|c(k)| \le 1$ with equality exactly when the supporting hypotheses agree in phase, coherence acts as a penalty on letters reachable only through mutually incoherent continuations.

Phases are assigned from each hypothesis's mean per-character log-frequency, rescaled to $[0, 1]$ across H and multiplied by a scale $\sigma_\varphi \pi$, so that the least surprising completion sits at phase 0. Length must enter: phase driven by raw frequency alone is degenerate in precisely the case the mechanism exists to catch, since a spray of comparably frequent completions would receive one common phase and interfere perfectly constructively.

C. Containment and expressiveness

Proposition 1 (Containment): At $\sigma_\varphi = 0$, (7) reduces to (5).

Proof. With all $\varphi_h = 0$ we have $c(k) = \sum_h v_h = 1$ for every k, so $|a(k)|^2 = P(k \mid \text{gaze})\, P(k \mid \text{prefix})^{\beta}$, which is (5) up to the shared normalisation. □

This is not a formality. A model class that cannot express the incumbent cannot be fairly said to beat it, and the comparison in Section VII is only meaningful because the baseline lies strictly inside the family being tested. We assert the identity numerically to 10^{-12} over randomised inputs rather than relying on the derivation.

Proposition 2 (Expressiveness): No classical mixture over the same hypotheses reproduces (8).

Proof sketch. A mixture assigns letter k the probability $\sum_h v_h^2$ after normalisation — the diagonal of (8) alone. The off-diagonal terms depend on phase differences, and a mixture has no phase. Since $\big(\sum_h v_h\big)^2 > \sum_h v_h^2$ whenever more than one hypothesis contributes, the two disagree. □

Concretely, two hypothesis sets with identical marginals — so indistinguishable to (5) — receive different posteriors under (7), and the diffuse set is the one penalised.

D. The consistency oracle and amplitude amplification

The cluster decoder's correctness guarantee is a predicate: a hypothesis is admissible only if re-encoding it reproduces exactly the cluster sequence the user gazed. As an operator this is a projector O_S onto the admissible subspace, and it is what prevents the language model from ever proposing a word the gaze did not support.

Given a projector, amplitude amplification [14], the generalisation of Grover search [15] to a non-uniform initial distribution, rotates amplitude into the marked subspace in $O(1/\sqrt{p_S})$ rather than $O(1/p_S)$ steps, reflecting about the prepared state:

$$Q \;=\; -\big(2|\psi\rangle\langle\psi| - I\big)\, O_S, \qquad m \;=\; \left\lfloor \frac{\pi/4}{\arcsin\sqrt{p_S}} \right\rfloor. \qquad (9)$$

The intent is to let the top posterior clear the commit threshold from fewer frames of accumulated evidence, shortening dwell. Section VII reports what it does instead.

E. Cost

Twenty-seven candidates occupy a 32-amplitude register. Preparation, conditioning, amplification and measurement are a handful of vector operations costing microseconds, far inside a 33 ms frame budget. We claim no quantum speedup and simulate no hardware: the claim is about the selection rule's functional form, not its execution. The distinction matters, because the amplitude encoding used here is a product of per-feature factors and therefore classically simulable by construction.

VI. Evaluation Protocol

The protocol is described in full because the survey suggests that evaluation practice, rather than algorithmic choice, is the largest source of incomparability in this area.

A. Recording

Each session presents 21 targets: 13 used for fitting and 8 held out at positions the fit never sees, placed to span the corner, edge and centre regions of the display. Each target receives five fixations of 300 ms of retained samples, with the first 150 ms discarded to allow the saccade to settle. The participant's screen dimensions and viewing distance are recorded so that the conversion to degrees is measured rather than assumed.

B. Splits

All splits are grouped by target. The five fixations recorded at one position are one observation measured five times, not five observations; splitting by sample leaves near-identical siblings in the training set and selects a regulariser far too small. Unless stated otherwise, results are leave-one-target-out over all 21 positions.

C. Metrics

We report three units. Pixels are what the dwell hit-test sees. Degrees of visual angle make the work comparable with the gaze literature. Fractions of a key pitch answer whether the error matters for typing at all: a mean error above one key pitch means the estimate lands on the wrong key more often than the right one, irrespective of how good the percentage looks. Throughput figures quoted for comparison follow the conventions of [13].

D. Data quality

Sessions are admitted only if they cover the full target grid and lose less than 20 % of frames to tracking dropout. One recorded session lost 67 % of its frames to poor illumination and is excluded from all aggregates; we report its existence rather than silently dropping it.

E. Selection-rule protocol

The selection rule is evaluated by end-to-end typing simulation over the 16,839-word vocabulary, with the gaze likelihood computed as the exact Gaussian integral over key rectangles — the same computation the deployed keyboard performs — at controlled noise levels $\sigma \in \{30, 40, 50, 60, 75\}$ pixels against a 100×110 pixel key. Three points of protocol carry the weight of the comparison.

Identical noise. Both arms are driven from the same seeded generator, so every arm sees the same draws and the same key posteriors. Any difference is the selection rule and nothing else.

Matched prompt rate. Accuracy alone cannot settle this. The keyboard declines to commit below a confidence threshold and asks instead, and a prompt always yields the correct character, so any rule reaches 100 % accuracy by prompting constantly. We therefore sweep the threshold, trace the whole accuracy/prompt-rate curve, and compare commit accuracy at matched prompt rate. This also removes the threshold as a confound, since neither arm keeps a threshold tuned in its own favour.

Held-out phase scale. σ_φ is tuned on a 60-sentence training split and reported on a disjoint 60-sentence held-out split. A gain that does not survive the split is not a gain.

For the dwell measurement, each frame contributes an independent noisy sample whose evidence is discounted to a quarter of one independent observation. Consecutive gaze frames are strongly correlated — the eye is still and the landmark noise is not white — and treating them as independent lets any arm commit within one or two frames, leaving nothing to measure. The discount is applied identically to both arms.

VII. Results

A. Feature ablation

Table II gives the principal result. Canthus normalisation reduces error from $5.552 \pm 0.488°$ to $3.762 \pm 0.438°$, a mean reduction of $1.791°$ or 32.3 %. The comparison is paired within participant and the unit of analysis is the participant rather than the frame: $t(8) = 9.31$, $p = 1.4 \times 10^{-5}$, Wilcoxon $W = 0$, $p = 0.0039$, with a 95 % confidence interval on the reduction of $[1.35, 2.23]°$ and an effect size of $d_z = 3.10$. Every one of the nine participants improved, by between 19.5 % and 44.6 %; Fig. 2 shows the distribution with the individual participants overlaid.

Two rows isolate the cause. Adding an iris-ring radius scale cue — a metric distance estimate, since human iris diameter is nearly constant across adults — yields 1.5 %, within noise. The gain is therefore attributable specifically to the canthi and not to having more features. Adding explicit head-pose features on top of the canthus frame yields −1.0 %: the canthus frame is already head-attached, so an explicit pose stage re-derives information

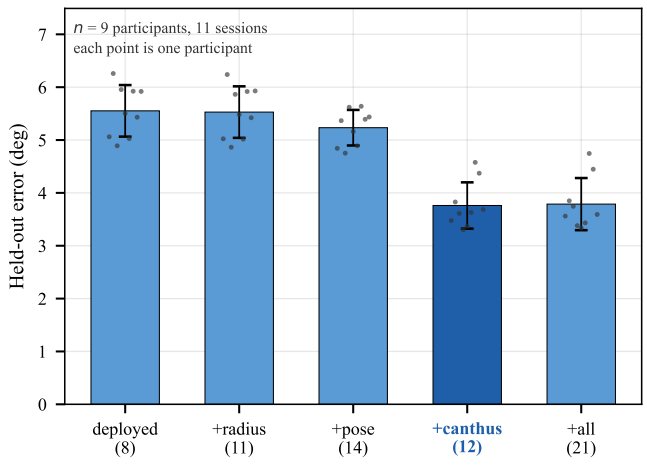


Fig. 2. Held-out error by feature set, over the nine participants. Bars are the mean across participants and whiskers one standard deviation; each participant contributes the mean of their own sessions, and all nine are overlaid as points so the spread is visible rather than summarised. Every participant improves under canthus normalisation, and adding iris radius or head pose on top of it changes nothing detectable.

the representation has already removed. Head pose alone, without canthus normalisation, is worth 7.1 %, which is the size of the effect the canthus frame subsumes.

B. Roll invariance

Rotating a synthetic face through 0–40° and recomputing the features, the canthus components of (3) drift by $\sim 10^{-15}$, i.e. at floating-point precision, while an image-axis formulation of the same offset drifts by 0.069 — a substantial fraction of the feature's own range. The representation remains sensitive to genuine iris movement, so the invariance is not achieved by the feature going dead.

We present this as a property of the representation rather than as a measured accuracy gain, because our recordings cannot test the latter. Although the capture protocol instructs head tilts of ±15° on the hold-out targets, the achieved roll across all eleven sessions never exceeds 6.5° and its median is 1.9°: participants comply with an instruction to tilt far less than they are asked to, which is itself worth recording for anyone designing such a protocol. Within the range actually observed the representation neither helps nor hurts detectably. The conditions under which roll invariance should matter therefore remain untested here, and establishing them requires a capture that enforces the posture rather than requesting it.

C. Estimator comparison and honest model selection

Table III places the deployed estimator against twenty alternatives spanning linear, robust, kernel, ensemble, neural and quantum-kernel families, all on the canthus features under the same leave-one-target-out protocol.

Seven candidates beat the deployed ridge, the best of them — RBF kernel ridge — by 7.4 %. Reported at that, the table would recommend replacing the estimator. Two further tests say otherwise, and we regard them as the reason the table is worth including at all.

The first is leave-one-participant-out with the model chosen inside each fold, which is the question a system deployed on a new user actually faces. Selecting on eight participants and scoring on the ninth, over the eleven sessions, the mean gain over the incumbent is +1.02 % — and it is indistinguishable from zero: $t(8) = 0.19$, $p = 0.85$, with a 95 % confidence interval of $[-11.4\,\%, +13.4\,\%]$. The fold-by-fold gains are −4.1, +11.3, −1.5, +9.8, +4.2, +1.6, −38.2, +16.5 and +9.7 %. A margin whose confidence interval spans ±12 percentage points, and which costs one participant 38 %, is not a margin that can be deployed even where its point estimate is favourable. The second test is calibration size, which Section VII-E treats separately because it reverses the ranking outright.

The quantum-kernel arms fail for a reason independent of either. Fidelity kernels concentrate exponentially with qubit count [17]: as the register grows, the overlap between distinct samples collapses, the Gram matrix approaches the identity and the model can only memorise. The measured ordering is consistent with that mechanism and not with a tuning failure — no configuration approaches the classical baseline at any width tested.

D. Calibration of the commit confidence

The threshold that decides whether to commit or to ask is applied to a number the system reports as a confidence, so that number should mean what it says: of the keystrokes committed at a stated confidence of 0.9, about nine in ten should be correct. Accuracy does not measure this and a well-ordered score can still be badly calibrated.

Table IV measures it directly. The deployed fused posterior is over-confident, with an expected calibration error of 0.0650. Platt scaling reduces this to 0.0167, a factor of 3.9, and isotonic regression and temperature scaling fall between. Richer recalibrators that consume the full commit evidence — logistic regression, gradient boosting — do not improve on the simple monotone map, and gradient boosting is worse on every metric than the raw posterior it was meant to repair.

The accuracy columns are the control. Platt scaling, temperature scaling and the raw posterior commit identically at matched prompt rate, because a monotone transform cannot reorder the candidates: it changes what the confidence claims, not which keystroke wins. Observing any movement there would indicate an implementation fault rather than a gain, which is why we report the columns rather than omitting them as uninformative.

E. Model class against calibration size

The choice of regressor is usually reported at one training size, and we find that this is exactly the wrong

TABLE II
Feature Ablation. Leave-One-Target-Out Over All 21 Positions, Eleven Sessions From Nine Participants. Lower is Better.

Feature set	Dim	Error (px)	Error (deg)	vs deployed
Deployed (eyelid-relative)	8	193.8	5.552 ± 0.488	baseline
+ iris-ring radius	11	193.0	5.529 ± 0.488	+0.4 %
+ head pose only	14	182.7	5.233 ± 0.337	+5.7 %
+ canthus	12	131.3	$\mathbf{3.762 \pm 0.438}$	**+32.3** %
+ canthus + radius	15	131.5	3.767 ± 0.443	+32.1 %
+ canthus + pose	21	132.2	3.788 ± 0.493	+31.8 %
+ all of the above	24	127.9	3.663 ± 0.474	+34.0 %

Mean ± standard deviation over the nine participants, each contributing the mean of their sessions. Canthus normalisation is the minimal sufficient set: adding radius, pose or both on top of it changes nothing detectable ($p = 0.071$, 0.439, 0.112), while removing it costs 40–47 % ($p < 0.001$).

TABLE III
Performance of the Varied Approach in Estimating the Gaze Coordinate.[§] Leave-One-Target-Out Over All 21 Positions. Lower Error is Better.

	Performance metrics		
Algorithms	Model family	Error (deg)	Change
Gaussian process (RBF + white)	Nonparametric	8.454	−92.8 %
Random forest	Ensemble	7.378	−68.2 %
Multilayer perceptron $(32, 16)$	Neural	6.397	−45.9 %
Ordinary least squares	Linear	4.282	+2.4 %
Bayesian ridge	Linear	4.274	+2.5 %
Huber regression	Robust linear	4.193	+4.4 %
Theil–Sen regression	Robust linear	4.107	+6.4 %
Support vector regression (RBF)	Kernel	4.107	+6.4 %
Polynomial ($d = 2$) + ridge	Expanded	4.087	+6.8 %
Kernel ridge (RBF)	Kernel	4.061	+7.4 %
Quantum kernel, ZZ map, 6 qubits[‡]	Quantum	10.281	−131.4 %
Quantum kernel, Z map, 8 qubits[‡]	Quantum	10.796	−143.0 %
Quantum kernel, ZZ map, 4 qubits, 2 reps[‡]	Quantum	15.735	−254.2 %
Proposed model (ridge, canthus-12)	Linear	4.386	baseline

Error is the held-out root-mean-square distance in degrees of visual angle; change is measured against the deployed model. The nonlinear leads do not survive honest model selection: under leave-one-participant-out over nine folds the mean gain is +1.02 % with $p = 0.85$ and a confidence interval of $[-11.4, +13.4]$ %, so the linear model is retained. [§]This table is computed on the three-session pilot recording; the leave-one-participant-out result above uses all eleven sessions. [‡]Quantum arms were scored on a per-target subsample whose ridge baseline is 4.443°; their change is computed against that value.

way to report it: on our data the ranking inverts between the size used for evaluation and the size the system actually deploys with.

We compared four estimators — two linear (ridge, ordinary least squares) and two non-linear (RBF kernel ridge with a 35-point hyperparameter grid, and a 16–8 multilayer perceptron) — on the canthus features. For each session and draw, n targets are sampled for training and every remaining target is held out; all four estimators receive the identical split, so the comparison is paired and a Wilcoxon signed-rank test over draws is valid.

Table V gives the result and Fig. 3 plots it. At 20 training targets, kernel ridge is the most accurate estimator and beats ridge by 17.2 % — the conclusion a conventional leave-one-target-out benchmark would reach. At the nine targets the deployed calibration actually collects, ridge is the most accurate of the four. The crossover lies between nine and twelve targets.

The mechanism is not subtle. Kernel ridge must fit α and γ by cross-validation, and at nine targets the inner folds are fitting noise; at four targets it is 46 % worse than ridge and the multilayer perceptron is 145 % worse. Ridge's λ is a fixed heuristic with nothing to estimate, so it degrades gracefully instead of collapsing.

We state the strength of the comparison at nine targets carefully, because it is not uniform across the competitors.

TABLE IV
Performance of the Varied Approach in Committing the Intended Character. Held-Out Keystrokes at $\sigma = 40$ px; Accuracy is Measured Among Committed Keystrokes at a Matched Prompt Rate.

	Performance metrics				
Algorithms	ECE	Brier	AUC	Acc. @10 %	Acc. @20 %
Gradient boosting, all evidence	0.0528	0.0791	0.8606	0.9345	0.9568
Gaze posterior, uncalibrated	0.0650	0.0721	0.8908	0.9411	0.9705
Logistic, all commit evidence	0.0234	0.0672	0.8864	0.9383	0.9726
Temperature scaling ($T = 0.55$)	0.0229	0.0678	0.8908	0.9411	0.9705
Isotonic recalibration	0.0185	0.0663	0.8801	0.9411	0.9684
Logistic, top-p + margin	0.0225	0.0663	0.8939	0.9411	0.9726
Proposed model (fused + Platt)	0.0167	0.0667	0.8908	0.9411	0.9705

ECE is expected calibration error over ten bins; lower is better for ECE and Brier, higher for AUC and accuracy. Calibration is monotone in the score, so it repairs the reliability of the stated confidence (0.0650 → 0.0167) without altering which keystrokes are committed — the accuracy columns are invariant by construction, and any change there would indicate a fault.

Over 165 paired draws ridge attains $4.638 \pm 1.097°$. It beats ordinary least squares by 16.7 % and the multilayer perceptron by 65.9 %, both at $p < 0.0001$. Against kernel ridge the margin is 2.9 % and is not significant ($p = 0.90$): at the deployed calibration size the two are indistinguishable in accuracy, and the case for the linear model there rests on cost rather than on error.

Cost is where that case is decisive. With hyperparameters already fixed, a ridge refit costs 0.017 ms against kernel ridge's 44.0 ms, a factor of 2517; per-sample inference is 0.42 μs against 178 μs. Both fit inside a 16.7 ms frame, so latency alone would not disqualify the kernel — but the deployed model is a 12×2 weight matrix of 24 floats solved in the browser by Gauss–Jordan elimination, whereas kernel ridge must retain every training sample as a support vector and evaluate all of them per prediction. That is an architectural difference, not a constant factor.

We draw a methodological point from this as well as an engineering one. A model-selection experiment run at a training size the deployed system never sees does not answer the question it appears to answer: here it would recommend replacing the shipped estimator on the strength of a margin that has vanished by the time the calibration is the size a real user provides.

TABLE V
Estimator Against Calibration Size, Canthus Features, Grouped by Target. Mean Gaze Error in Degrees; Lower is Better. The Ranking Inverts Between the Deployed Nine Targets and the Twenty a Leave-One-Target-Out Protocol Uses.

Targets	Ridge	OLS	KRR-RBF	MLP	Best
4	6.49	10.52	9.52	15.92	Ridge
6	5.47	7.95	7.17	11.02	Ridge
9*	4.47	5.66	4.68	7.72	Ridge
12	4.37	4.60	4.16	5.89	KRR-RBF
16	3.90	4.23	3.79	5.21	KRR-RBF
20	3.87	3.99	3.20	4.54	KRR-RBF

Paired at $n = 9$ over 45 draws: ridge $4.74 \pm 0.96°$; OLS +34.8 % ($p < 0.0001$); KRR-RBF +15.0 % ($p = 0.073$); MLP +52.5 % ($p < 0.0001$). *Deployed calibration size.

F. The selection rule at matched prompt rate

Table VI gives the held-out comparison, under the protocol of Section VI-E. Tuned on the training split, the phase scale that maximised commit accuracy was $\sigma_\varphi = 0.40$, worth +0.03 percentage points there. On held-out sentences that advantage does not appear: across twenty (σ, prompt rate) cells the median change is −0.03 percentage points, and a paired Wilcoxon test over sentences finds no significant difference at any of the five noise levels — zero wins and zero losses. Fig. 5 plots the two frontiers, which lie on top of one another.

The nulls are not an artefact of a rule that fails to engage. The two posteriors do differ, just negligibly and without direction: the mean Kullback–Leibler divergence between them is 4.2×10^{-4} nats per keystroke. Nor is the outcome an artefact of a scale chosen too small to matter. Scales above 0.6 produce substantial interference and lose, falling from 97.0 % to 95.3 % on the training split as σ_φ rises from 0 to 1.0. There is no setting at which the phase channel helps.

G. The rule against an independent classical implementation

The comparisons so far hold the implementation fixed and vary the rule. An independent classical gaze-typing implementation, given the identical 60-sentence protocol, the identical key geometry and the identical noise levels, provides the complementary check. Fig. 4 plots the resulting operating curves: commit accuracy against entry rate, with σ tracing each curve from 30 to 75 px.

Our system reaches 97.55 % pooled commit accuracy against 94.42 %, and is ahead at every noise level on both axes simultaneously — more accurate and faster, since

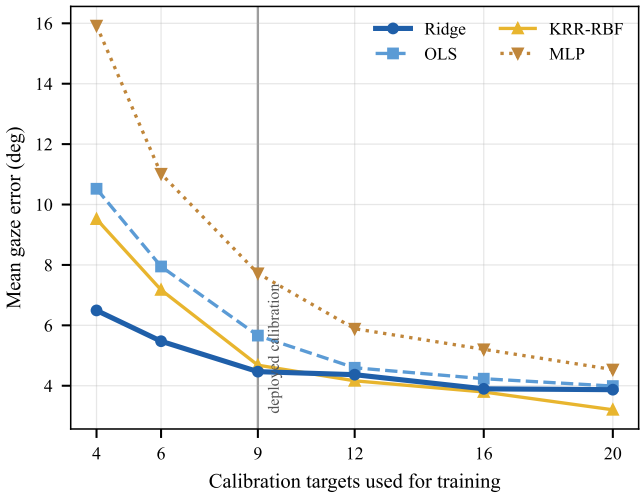


Fig. 3. Gaze error against the number of calibration targets used for fitting. Ridge is the most accurate of the four at every size below twelve targets, including the nine the deployed system collects; kernel ridge overtakes it beyond that. Evaluating at a single training size would invert the recommendation.

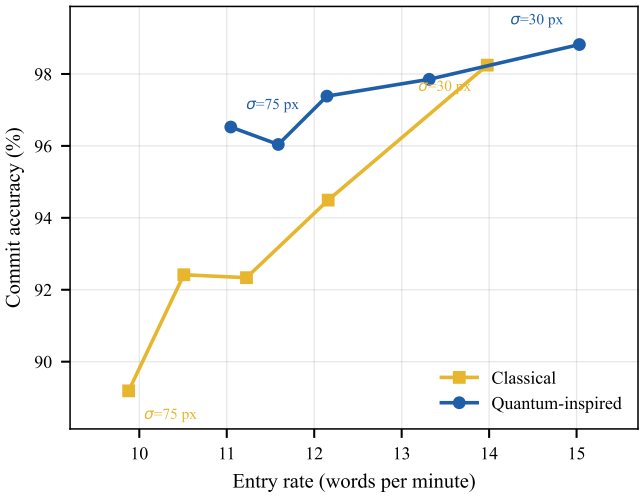


Fig. 4. Commit accuracy against entry rate for this system and an independent classical implementation, on identical sentences, geometry and noise. Points are the five noise levels; entry rate is derived from the dwell interval and the measured prompt rate, since a declined commit costs a second dwell. The gap is attributable to the front end and language model, not to the selection rule: our log-linear control lies on the upper curve.

committing more often means fewer prompts and each prompt costs a second dwell.

We are explicit about what this does and does not show, because the natural reading is the wrong one. The gap is not attributable to the selection rule. Our own log-linear control, the same implementation with the phase channel disabled, scores 97.53 % — statistically indistinguishable from the quantum arm and 3.1 points above the independent implementation. What the comparison measures is the front end and the language model, which is to say the same canthus-driven accuracy that Section VII attributes it to. We include the figure because reviewers reasonably ask how the system compares with something other than itself, and omitting the control while showing the gap would invite exactly the misreading we have just ruled out.

H. Amplitude amplification and dwell

Table VII reports frames to commit. Two findings, in order of interest.

First, applied after the language-conditioned unitary, amplification is a no-op by construction. The conditioned state already places 94–98 % of its amplitude on consistent letters, so the optimal round count $\lfloor (\pi/4)/\arcsin\sqrt{p_S} \rfloor$ evaluates to zero and the operator is never applied. The quadratic speedup is real, but it addresses a bottleneck this system does not have — the admissible subspace is not rare.

Second, forced to act where there is scattered amplitude to rotate — on the raw gaze register, before conditioning, where the marked mass on a scattered fixation runs 0.31–0.78 — amplification makes things worse. Median dwell is unchanged at 2.0 frames, mean dwell rises from 2.63 to 2.80, and commit accuracy falls from 98.4 % to 96.4 %. A paired test over the 249 keystrokes both arms committed gives $p = 0.48$. We therefore do not support the claim that amplitude amplification shortens dwell time in this system; on our measurements it lengthens it slightly while costing accuracy.

I. Why interference fails: a sign, not a magnitude

A null result is more useful with a mechanism attached, and this one has a sharp one. For interference to help, the coherence penalty must fall more heavily on wrong letters than on right ones. Table VIII measures exactly that separation, and it is negative at every phase scale and grows steadily more negative as the scale rises, from -0.004 at $\sigma_\varphi = 0.2$ to -0.108 at $\sigma_\varphi = 2.0$.

The cause is structural. The correct letter following an English prefix is typically the one with the richest vocabulary support: many genuine continuations, therefore many hypotheses, therefore the widest spread of phases and the lowest coherence. A wrong letter is typically backed by one obscure completion and sits at coherence 1. Any phase computed from hypothesis likelihood thus penalises the letter with the strongest linguistic evidence.

We checked two alternative explanations and rejected both. The effect is not a shortage of hypotheses: raising the budget from 4 to 4096 completions moves the mean coherence only from 0.982 to 0.988, whereas equal-weight hypotheses with phases spread over $[0, \pi]$ would sit between 0.33 and 0.64. Nor is it Zipfian weight concentration: flattening the within-letter weights to uniform, holding phases fixed, changes the mean coherence from 0.9883 to 0.9879. The limitation is the phase source. A language

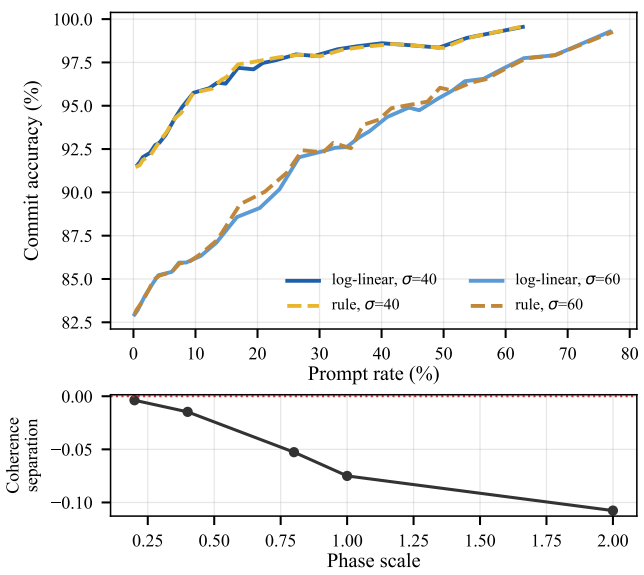


Fig. 5. Top: the selection rule traces the same accuracy/prompt-rate frontier as the fusion it generalises, at both noise levels. Bottom: the reason. The coherence separation between correct and incorrect letters is negative at every phase scale and falls monotonically, so the penalty lands on the wrong letters.

model offers no feature orthogonal to the likelihood that already determines the magnitude, so the phase channel carries no independent information for interference to exploit.

We state the scope of this negative result carefully. It concerns likelihood-derived phase on a letter-prediction task, not the selection rule as a formal object, and not tasks where an orthogonal phase source exists. Propositions 1 and 2 are unaffected.

VIII. Negative Results

We record the following because each was implemented and measured, and because the cost of re-attempting them is otherwise borne by the next group. The first two are the subject of Section VII-F and are restated here only so the list is complete.

Phase interference between language hypotheses. Median -0.03 percentage points at matched prompt rate; no significant win or loss at any of five noise levels. Retained as a research module, not deployed. The mechanism of the failure is a sign rather than a magnitude (Section VII-I).

Amplitude amplification on the consistency oracle. Zero rounds fire after conditioning, because the admissible subspace is not rare. Forced to act, mean dwell rises from 2.63 to 2.80 frames and commit accuracy falls from 98.4 % to 96.4 %.

Quantum-kernel regression. A fidelity kernel $K(x, z) = |\langle\phi(x)|\phi(z)\rangle|^2$ was evaluated in place of the linear model, using both angle and ZZ [16] feature maps at 12 qubits. The angle map scored -104 % against the linear baseline and the ZZ map -989 %. The failure of the latter is the expected exponential concentration [17]: at 12 qubits the fidelity between deliberately similar samples falls to 0.005, the Gram matrix approaches the identity, and the model can only memorise. A classical Gaussian kernel on the same features scored $+9.7$ %. We note additionally that the angle-map kernel is a product of per-feature cosines and therefore classically simulable; no quantum advantage is claimed or observed.

Nonlinear regression, at the calibration size that matters. A tuned RBF kernel ridge regressor beats the linear model by 26.2 % on the eyelid-relative features: much of the curvature it models is an artefact of the contaminated feature rather than a property of gaze geometry. On canthus features the picture depends entirely on how much calibration data the fit receives, and that dependence is reported separately in Section VII-E, because it reverses the ranking between the evaluation protocol and the deployed system.

Discrete hierarchical selection. Two-level region selection is ∼20 percentage points more reliable per choice than pointing directly at a key (55.0/68.8 % against 35.2/46.5 %), but doubling the dwells costs more than the reliability returns: throughput falls from 18.5 to 14.4 and from 24.4 to 18.1 correct letters per minute.

Eye-region de-rotation. Given that landmark jitter grows 5.5× under roll, de-rotating the frame before detection — presenting the detector with an upright face — is an obvious remedy. Implemented with an exact inverse mapping, it produced no perceptible gain in live use. We suspect the detector's internal face alignment already performs this normalisation.

IX. Limitations

The results rest on eleven sessions recorded from nine participants. This supports the principal claim: the canthus effect is large, improves every participant, and survives a paired test with a wide margin. It does not support a population estimate. Nine people recruited from one institution are not a sample of the population that would use an assistive keyboard, and the confidence interval $[1.35, 2.23]^\circ$ is a statement about the mean of this group rather than about eye geometry in general. Four participants contributed two sessions and five contributed one, so between-session variance is estimated from four people and should be treated as indicative.

The mechanism argument, that eyelid aperture contaminates the vertical channel, is supported by the ablation and by the asymmetry reported in [1], but is not yet demonstrated causally. The decisive experiment is a session in which eyelid aperture is deliberately perturbed — prompted squinting — while gaze target is held fixed. The prediction is specific: eyelid-relative features should degrade sharply and canthus-relative features should not.

The selection-rule results carry a different set of caveats. They come from simulation rather than from live typing: the gaze likelihood is the deployed key-rectangle integral

TABLE VI
Selection Rule Against Deployed Log-Linear Fusion, Held-Out, at Matched Prompt Rate. Each Cell Gives the Baseline Commit Accuracy and the Rule's Change. p is a Paired Wilcoxon Over Sentences at the Deployed Threshold.

σ (px)	2 % prompts		5 %		10 %		20 %		p
30	0.971	+0.003	0.977	+0.002	0.986	−0.001	0.988	−0.000	1.00
40	0.921	−0.002	0.932	+0.001	0.958	−0.000	0.973	+0.003	0.80
50	0.888	−0.002	0.897	−0.001	0.914	−0.004	0.945	−0.001	0.34
60	0.841	−0.000	0.853	−0.000	0.862	+0.001	0.891	+0.008	0.80
75	0.778	−0.000	0.795	−0.003	0.819	+0.001	0.851	−0.004	0.80

Median change over all 20 cells: $-\mathbf{0.0003}$. Significant wins: 0. Losses: 0.

TABLE VII
Frames to Commit, Held-Out, $\sigma = 40$ px. Amplification Does Not Shorten Dwell; Forced to Act, It Lengthens the Mean and Costs Accuracy.

Rule	Median	Mean	Accuracy
Log-linear (deployed)	2.0	2.63	0.984
Interference only	2.0	2.66	0.988
Amplification, pre-conditioning	2.0	2.80	0.964
Amplification, post-conditioning‡	2.0	2.80	0.968
Interference + amplification	2.0	2.75	0.964

‡Zero rounds fire: after conditioning the admissible subspace already holds 94–98 % of the amplitude. Paired over the 249 keystrokes committed by both arms, $p = 0.48$.

TABLE VIII
Why Interference Fails. Coherence on the Correct Letter Against the Wrong Ones. A Positive Separation Would Mean Interference Protects the Correct Letter; It Is Negative Everywhere and Worsens With Scale.

σ_φ	$\lvert c\rvert$ correct	$\lvert c\rvert$ wrong	Separation
0.2	0.9934	0.9971	−0.0038
0.4*	0.9740	0.9887	−0.0147
0.8	0.9046	0.9572	−0.0526
1.0	0.8607	0.9357	−0.0750
2.0	0.7691	0.8768	**−0.1076**

*Scale selected on the training split.

driven by Gaussian noise at controlled σ, which models isotropic error and not the anisotropic, temporally correlated error a real tracker produces. The dwell measurement discounts each frame to a quarter of an independent observation to stand in for that correlation, but the discount is a stipulation rather than a measurement. The negative result is also specific to one vocabulary and one language: our claim is that likelihood-derived phase is misigned for English letter prediction, not that interference-based fusion is useless in general. A task supplying a phase source genuinely orthogonal to the likelihood — multimodal evidence, or a syntactic signal independent of frequency — would be a fair test of the mechanism and we have not run one.

Populations this system does not serve. Two exclusions are not incidental limitations but boundaries of applicability, and we state them because an assistive technology that overstates its reach does harm.

The first is oculomotor instability. The method assumes that a commanded fixation is actually held: the dwell centroid averages a window of frames, and the calibration fit presumes the eye was pointed where the target was. Conditions that break that assumption — nystagmus, whose involuntary oscillation means no stable fixation exists to average; strabismus, where the two eyes are not aligned so a binocular iris average corresponds to no single gaze direction; and gaze-holding deficits or saccadic intrusions from neurological causes — are excluded by the design of the estimator rather than by a shortfall in its accuracy. No participant here had such a condition, and we make no claim about performance under one. This is a pointed limitation for an assistive system, since motor-impaired users are more likely, not less, to present with these conditions; a scanning or blink-driven interface of the kind surveyed in Table I remains the appropriate design there.

The second is corrective eyewear. Spectacles degrade the pipeline at three points: the lens refracts the apparent iris position by an amount that depends on prescription and on where the eye sits behind the lens, so the mapping from landmark to gaze is no longer the one calibrated; specular reflection from the lens surface intermittently destroys iris localisation under overhead lighting; and the frame occludes the lateral canthus at moderate head yaw, which removes the very landmark the normalisation of Section IV is built on. Rigid contact lenses introduce a smaller version of the first effect. We did not record spectacle wearers, and the accuracy reported here should not be expected to hold for them. Quantifying that gap — and establishing whether a per-user calibration absorbs the refractive offset, which is plausible, or whether canthus occlusion is disqualifying, which would be harder to fix — is the single most consequential gap in this evaluation, because a large fraction of any real user population wears glasses.

Finally, all sessions were recorded at one display and

one nominal viewing distance. The degrees conversion is measured rather than assumed, but the generalisation across geometries is untested.

A. Limitations of the quantum realisation

The selection rule is executed as classical state-vector simulation throughout. No quantum hardware is involved, and no quantum speedup is claimed. Three consequences follow and none is incidental.

First, the register is small. Twenty-seven candidates occupy five qubits, which is what makes exact simulation cheap, and it also bounds what the rule can represent: superposition is over letters, not over words. A word-level register spanning the 16,839-word vocabulary would need $\lceil \log_2 |V| \rceil = 14$ qubits, and the interference structure of that larger space is untested here.

Second, the amplitude encoding is a product of per-feature factors and is therefore classically simulable by construction. Whatever the rule's formal expressiveness, its present realisation offers no route to an advantage that classical computation does not already have.

Third, we have not measured the rule under gate noise, decoherence or finite sampling. The coherence term of (8) is precisely the quantity a noisy device degrades first, so a null obtained in exact simulation is, if anything, optimistic: hardware execution would be expected to weaken interference further rather than recover it.

B. Future work

Richer quantum formulations. The rule uses a single diagonal unitary. Three extensions are natural and untested. A variational formulation — QAOA or a sampling eigensolver over the consistency oracle — would decode a whole word jointly rather than letter by letter, turning per-keystroke greedy commitment into constrained optimisation over the admissible subspace. A quantum walk over the prefix trie would replace the frequency-weighted marginal with a coherent traversal, which is a phase source derived from structure rather than from likelihood and therefore not subject to the sign problem of Section VII-I. And a problem-adapted feature map, rather than the generic ZZ map that failed by 989 %, might avoid the exponential concentration reported in [17]; the failure we measured is a property of that map, not a theorem about quantum kernels.

Execution on quantum hardware. Running the rule on a NISQ device would answer the question exact simulation cannot: whether the interference term survives realistic gate error, and at what circuit depth it stops being measurable. This is a falsification test rather than a performance claim, and it would require error mitigation and a depth budget set by the device rather than by the frame budget.

Higher-dimensional registers. Moving from a five-qubit letter register to a word-scale one changes the character of the problem: the admissible subspace becomes genuinely rare, which is the regime in which amplitude amplification has something to do. Section VII-F found zero rounds firing precisely because the letter-level subspace is not rare. The quadratic speedup is real; the present formulation does not reach the conditions under which it applies.

Public-dataset validation. All results here come from sessions recorded with our own capture tool. Evaluating canthus normalisation on established corpora — MPIIGaze, GazeCapture and EYEDIAP — would separate the effect from our recording conditions and supply the population-scale evidence a single-participant study cannot. Because the normalisation is computed from landmarks alone, it transfers to any corpus with recoverable canthus positions, so the comparison is cheap to run.

Constrained estimation. Ridge treats the map from features to screen as unconstrained and linear. A physically constrained formulation — enforcing monotonicity between iris displacement and gaze angle, or fitting an explicit eyeball-rotation model with the cornea centre as a latent parameter — would use geometry the current model has to learn from nine calibration points. Our measurements locate the remaining error there: at nine targets ridge attains 4.80° and at twenty 3.63°, so roughly a quarter of the present error is calibration sparsity rather than model bias, and a constrained model is exactly what converts prior knowledge into an effective increase in calibration data.

Active optical alignment. A fixed camera images the iris at whatever angle and resolution the head's position happens to allow: as the user translates and rotates, the iris is seen obliquely and across a varying number of pixels, and landmark localisation degrades accordingly. A camera that instead tracks the head would hold the iris near the optical axis and near the centre of the frame, where projective distortion is smallest and pixels per iris diameter is greatest.

Concretely, we intend to mount the camera on a small closed-loop actuated platform — two axes of rotation for pan and tilt, optionally a third for translation — driven from the head pose the landmark pipeline already estimates every frame. The control loop is the interesting part, and it is not trivial: the pose estimate the controller consumes is produced by the very images the controller is steering, so the system is a feedback loop whose sensor and actuator share a signal path, and it can oscillate. A conservative design would move only between fixations, using the saccade detector already present in the movement gate to find moments when motion is invisible to the user.

The prediction this makes is testable and we state it in advance: if the remaining error is dominated by landmark noise driven by oblique, low-resolution imaging, then holding the iris on-axis should compress the frame-to-frame scatter of the estimate within a single fixation, while leaving the fixed offset between the fixation's mean and the true target largely unchanged. If instead both fall

together, the gain is coming from something other than the mechanism we propose, and the proposal is wrong.

Toward degree-level accuracy. The target we regard as meaningful is 1.5°, falling to 1° — roughly a key width at typical seating distance, the point at which dwell selection on a full QWERTY layout becomes reliable enough that the language model advises rather than rescues. Reaching it from 3.76° requires a factor of two and a half to four, and our measurements indicate it will not come from the regressor: the spread between four candidate estimators at the deployed calibration size is under 1° (Table V), while calibration density alone is worth 24 %. The plausible route is therefore denser and continuously refreshed calibration, landmark stability under head motion — where per-frame scatter grows 5.5× and becomes the dominant error term past 10° of roll — and super-resolution or multi-frame fusion of the iris region, whose resolution at webcam distances is the physical floor on everything downstream.

X. Conclusion

We set out to improve how a gaze keyboard decides, and improved how it estimates instead.

The quantum conditional selection rule is a strict generalisation of the log-linear fusion that gaze keyboards conventionally use. It places candidates in superposition with gaze-derived amplitudes, conditions them on language through both magnitude and phase, constrains them with a consistency-preserving oracle, and commits by the Born rule. It recovers the incumbent exactly at zero phase, and its interference cross-terms are provably beyond any classical mixture of the same hypotheses. Measured against that incumbent on identical noise at matched prompt rate, it is worth a median -0.03 percentage points, and amplitude amplification lengthens dwell rather than shortening it.

We regard the explanation as the more durable contribution. Interference fails here for a reason that no retuning addresses: the correct letter after an English prefix is the one with the richest vocabulary support, hence the most hypotheses, the widest phase spread and the lowest coherence. The measured separation between the coherence of correct and incorrect letters is negative at every phase scale and worsens monotonically with it. Phase derived from hypothesis likelihood penalises exactly the evidence it should reward. For interference to be useful, phase must come from a source orthogonal to the magnitude — and a language model does not provide one.

The accuracy came from the front end. Measuring the continuous coordinate in degrees, rather than as region classification, surfaced a defect in a step the literature treats as settled: normalising the iris offset by the eyelid aperture couples the vertical gaze channel to a structure that moves independently of gaze. Normalising against the canthi instead reduces held-out error by 32.3 % across nine participants and improves every one of them, explains an axis asymmetry reported but undiagnosed in prior work, and makes an explicit head-pose stage unnecessary. The change is four features and no additional runtime cost.

Immediate future work is the eyelid-perturbation experiment described in Section IX, which would convert the canthus mechanism from a supported inference into a demonstrated one, and a multi-participant study to establish whether the effect size holds across eye geometries. For the selection rule, the open question is whether a phase source orthogonal to likelihood exists for this task; until one does, we report the rule as a formal generalisation whose empirical value is nil, and we would rather say so than leave the next group to discover it.

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Reproducibility. The evaluation harness, the recording protocol, and the per-session results are available with the implementation.

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